

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 03 · Simultaneous Equations

Questions

Student book · questions with writing space

- 14 pages
- Past papers up to June 2026

Simultaneous Equations

Your classified route

5
questions

31
marks

2
families

4
variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Simultaneous Equations

Families and variants

F01 • Model-parameter Simultaneous Systems

V01 Two-observation Calibration and Prediction

F02 • Nonlinear Systems by Substitution

V01 Linear plus Quadratic System

V02 Polynomial Pair to a Biquadratic

V03 Rational-cubic Intersections with Follow-on Geometry

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Model-parameter Simultaneous Systems

WHAT THIS FAMILY ASKS

Use two observed data points to form and solve a linear simultaneous system in two model constants, then invert the calibrated model for a prediction.

VARIANT ROUTE

V01 Two-observation Calibration and Prediction

2 Complete Classified questions

11 total marks

2. A tree was planted.

Exactly 3 years after it was planted, the height of the tree was 2 m.

Exactly 5 years after it was planted, the height of the tree was 2.4 m.

Given that the height, H metres, of the tree, t years after it was planted, can be modelled by the equation

$$H^3 = pt^2 + q$$

where p and q are constants,

(a) find, to 3 significant figures where necessary, the value of p and the value of q .

(4)

2. A tree was planted.

Exactly 3 years after it was planted, the height of the tree was 2 m.

Exactly 5 years after it was planted, the height of the tree was 2.4 m.

Given that the height, H metres, of the tree, t years after it was planted, can be modelled by the equation

$$H^3 = pt^2 + q$$

where p and q are constants,

Exactly T years after the tree was planted, its height was 5 m.

(b) Find the value of T according to the model, giving your answer to one decimal place. (2)

3. The population of a town was monitored.

Exactly 5 years after monitoring began, the population was 58 000

Exactly 10 years after monitoring began, the population was 65 000

Given that the population of the town, P thousand, t years after monitoring began can be modelled by the equation

$$P^2 = a + bt^3$$

where a and b are constants,

(a) find the value of a and the value of b .

(3)

3. The population of a town was monitored.

Exactly 5 years after monitoring began, the population was 58 000

Exactly 10 years after monitoring began, the population was 65 000

Given that the population of the town, P thousand, t years after monitoring began can be modelled by the equation

$$P^2 = a + bt^3$$

where a and b are constants,

According to the model, exactly T years after monitoring began, the population was 85 000

Making your method clear,

(b) find the value of T , giving your answer to one decimal place.

(Solutions relying entirely on calculator technology are not acceptable.)

(2)

Nonlinear Systems by Substitution

WHAT THIS FAMILY ASKS

Isolate one variable or equate two curve expressions, reduce the system to a solvable quadratic structure, and back-substitute to obtain matched ordered pairs and any requested geometry.

VARIANT ROUTE

V01 Linear plus Quadratic System

V02 Polynomial Pair to a Biquadratic

V03 Rational-cubic Intersections with Follow-on Geometry

3 Complete Classified questions

20 total marks

2.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(a) Given

$$y - x = 4$$

and

$$2x^2 - xy = 8$$

show that

$$x^2 - 4x - 8 = 0$$

(2)

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$$x^2 - 4x - 8 = 0$$

(b) Hence solve the simultaneous equations

$$y - x = 4$$

$$2x^2 - xy = 8$$

writing the answers in the form $p + q\sqrt{3}$, where p and q are integers to be found.

(4)

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6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0$$

(2)

$$2x^4 - 15x^2 - 50 = 0$$

(b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$

$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

(5)

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8. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \quad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

(a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0$$

(2)

8.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \quad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

Given that C_1 and C_2 meet at points P and Q (b) find, using algebra, the exact distance PQ

(5)

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